

TOP 20 VIVA QUESTIONS — CRYPTOGRAPHY & NETWORK SECURITY LAB

(With Short, Simple Answers)

- 1. What is cryptography?** The science of protecting information by transforming it into an unreadable form (ciphertext) so only authorized parties can read it.
- 2. Difference between plaintext and ciphertext?** Plaintext is the original readable message; ciphertext is the scrambled, unreadable form produced after encryption.
- 3. What is the Caesar cipher?** A substitution cipher that shifts every letter of the alphabet by a fixed number of positions (the key, k).
- 4. Why is the Caesar cipher weak?** It has only 25 possible keys, so it can be broken instantly by brute-force (trying all shifts).
- 5. What is a monoalphabetic substitution cipher?** A cipher where each plaintext letter is mapped to one fixed, unique ciphertext letter using a permuted alphabet.
- 6. Why can monoalphabetic ciphers still be broken easily?** Because they preserve letter frequency patterns of the language, so frequency analysis reveals the mapping.
- 7. What is the Playfair cipher?** A digraph (two-letter-at-a-time) substitution cipher that uses a 5×5 matrix built from a keyword.
- 8. Why is Playfair more secure than simple substitution?** It encrypts pairs of letters instead of single letters, which hides single-letter frequency information.
- 9. What is a polyalphabetic cipher (Vigenère)?** A cipher that uses a different monoalphabetic substitution for each letter, driven by a repeating keyword.
- 10. What is the affine cipher?** A cipher of the form $C = (aP + b) \bmod 26$, combining multiplication and addition; requires $\gcd(a, 26) = 1$ to be reversible.
- 11. What is a one-time pad, and why is it unbreakable?** A cipher using a truly random key as long as the message, used only once — since any plaintext is equally likely, ciphertext gives no information.
- 12. What is the Hill cipher?** A polygraphic cipher that encrypts blocks of letters using matrix multiplication modulo 26.
- 13. What is a known-plaintext attack?** An attack where the cryptanalyst has matching pairs of plaintext and ciphertext and uses them to deduce the key.
- 14. What is DES?** Data Encryption Standard — a symmetric block cipher that encrypts 64-bit blocks using a 56-bit key over 16 Feistel rounds.
- 15. Why was DES considered insecure over time?** Its 56-bit key is too short — it can be brute-forced with modern computing power.
- 16. What is 3DES (Triple DES)?** Applying the DES algorithm three times with (typically two or three) different keys to increase effective key strength.

17. Difference between ECB and CBC mode? ECB encrypts each block independently (identical plaintext blocks give identical ciphertext); CBC XORs each plaintext block with the previous ciphertext block, so errors and patterns are chained/hidden.

18. What is S-DES? Simplified DES — a small, teaching version of DES using an 8-bit block and a 10-bit key, with the same structural ideas as full DES.

19. What is RSA, and why is it secure? An asymmetric (public-key) cipher based on the difficulty of factoring the product of two large prime numbers.

20. What is the Diffie-Hellman key exchange used for? It lets two parties agree on a shared secret key over an insecure channel without ever transmitting the key itself, based on the difficulty of the discrete logarithm problem.

Prepared as a quick revision sheet — read each answer aloud once before the viva; these cover the core concept behind every program in the lab list.